Basic Principles of Medicine 1

Module: Musculoskeletal System | DLA 12

An Overview of the Histopathological Aspects of the Skin

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Your Objectives will show here!

Objectives

SOM.MK.I.BPM1.2.MSK.3.HCB.1043	Identify the cell responsible for the development of melanoma and epidermal features listed in the diagnostic criteria.
SOM.MK.I.BPM1.2.MSK.3.HCB.1044	Discuss the 3 classifications of burns.
SOM.MK.I.BPM1.2.MSK.3.HCB.1045	List the 4 stages of wound healing.
SOM.MK.I.BPM1.2.MSK.3.HCB.1046	Describe the changes to the skin that results in pemphigus vulgaris.
SOM.MK.I.BPM1.2.MSK.3.HCB.1047	Describe skin pigmentation changes seen in vitiligo.
SOM.MK.I.BPM1.2.MSK.3.HCB.1048	Describe skin pigmentation changes seen in albinism.
SOM.MK.I.BPM1.2.MSK.3.HCB.1049	Describe the inflammatory changes seen in sebaceous glands in the clinical condition called acne.
SOM.MK.I.BPM1.2.MSK.3.HCB.1050	Describe the changes to the skin that results in bullous pemphigoid.
SOM.MK.I.BPM1.2.MSK.3.HCB.1051	Discuss keloid formation.
SOM.MK.I.BPM1.2.MSK.3.HCB.1052	Define acanthosis, acantholysis, hyperkeratosis and parakeratosis.
SOM.MK.I.BPM1.2.MSK.3.HCB.1053	Compare and contrast basal cell carcinoma and squamous cell carcinoma

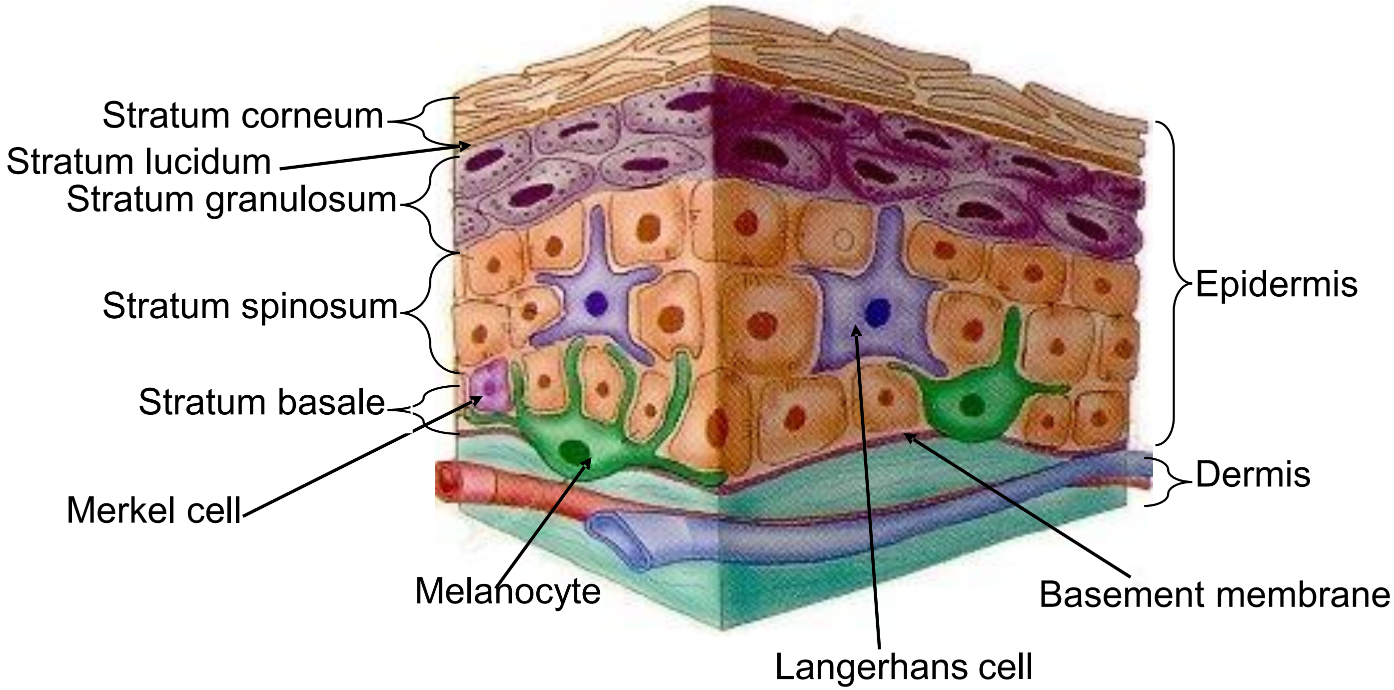
Recommended Reading

- Histology – A Text and Atlas, Pawlina 9th Edition
- Chapter 15: Blood Pg. 538-587

https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=257426085&bookid=3290

- Chapter 14: Textbook of Histology Pg. 333-354 (Read the section titled "Pathological Considerations"
- https://www.clinicalkey.com/student/content/book/3-s2.0-B978032367272600014X#hl0001002

Epidermis



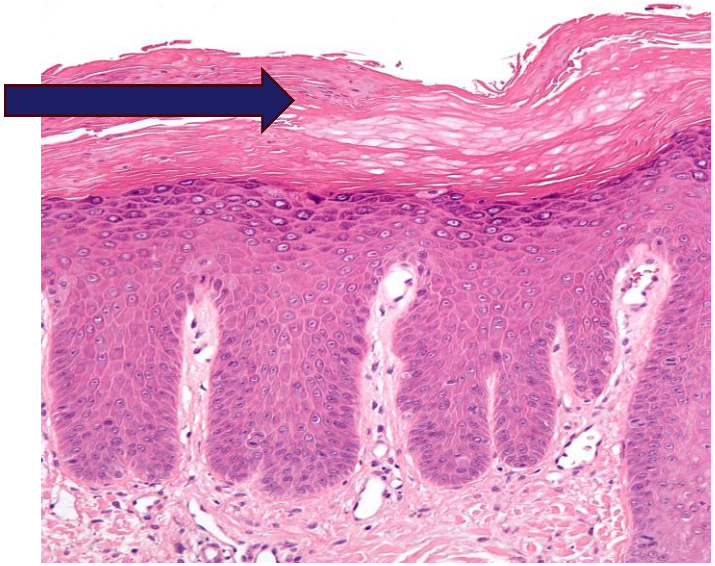
Epidermal Cells
- Keratinocytes
- Melanocytes
- Langerhans
- Merkel

Definition of Terms

Hyperkeratosis — thickening of the **stratum corneum** due to an excess production of keratin, a key structural protein in the skin.

Associated conditions:

1. Corns and Calluses: These are thickened areas of skin caused repeated friction or pressure
2. Warts: Caused by the human papillomavirus (HPV),
3. Psoriasis: This chronic autoimmune condition results in patches of thickened, scaly skin3
4. Actinic Keratosis: These are precancerous growths caused by long-term sun exposure
5. Keratosis Pilaris: "chicken skin," small, rough bumps, typically on the upper arms, thighs, or cheek

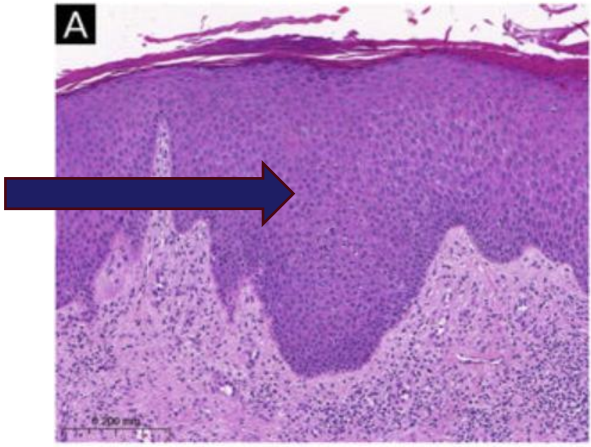


Definition of Terms

Acanthosis – increase in thickness of the **stratum spinosum**. Acanth" in acanthosis is derived from the Greek word "akantha" meaning "***thorn***" or "***prickle***" (spinous layer is also called the prickle cell layer) and "osis" refers to an "increase".

Associated Conditions

1. Acanthosis Nigricans: dark, velvety patches of skin found in body folds; commonly associated with insulin resistance, obesity, and sometimes malignancies
2. Psoriasis: chronic autoimmune condition
3. Chronic Eczema: long-term inflammation and irritation of skin
4. Seborrheic Keratosis: benign, wart-like lesions



Definition of Terms

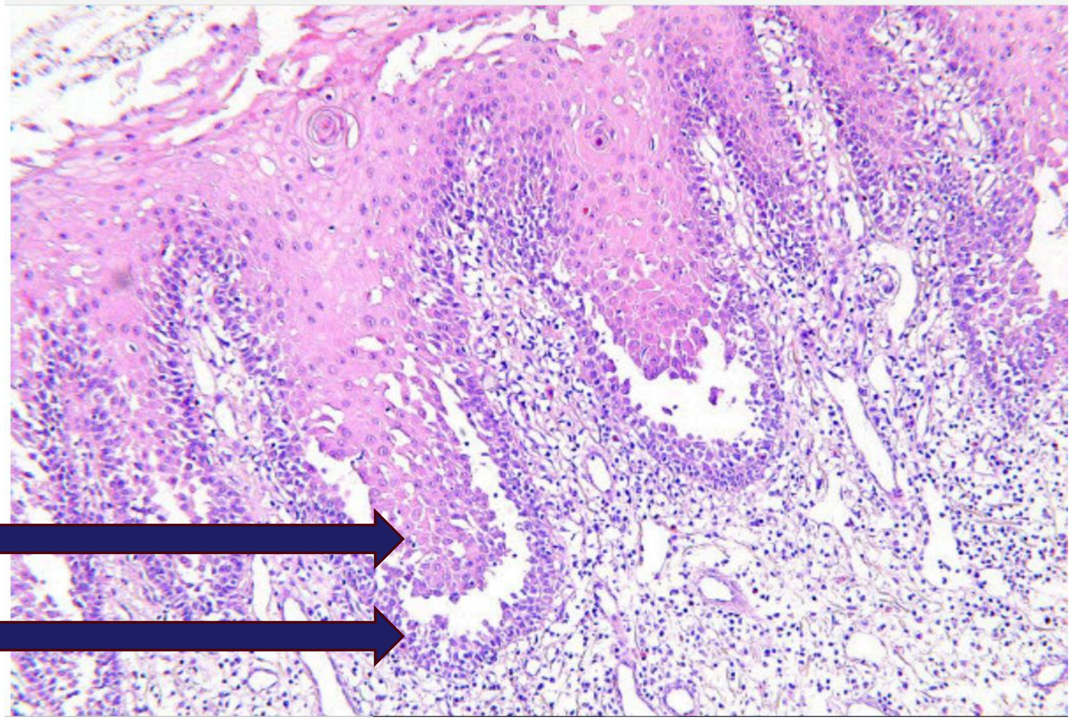
Acantholysis – broken bonds between cells of the epidermis leading to separation of these cells from each other

- Associated conditions: pemphigus vulgaris, varicella zoster virus infections, keratosis follicularis

Chicken Pox

Stratum Spinosum

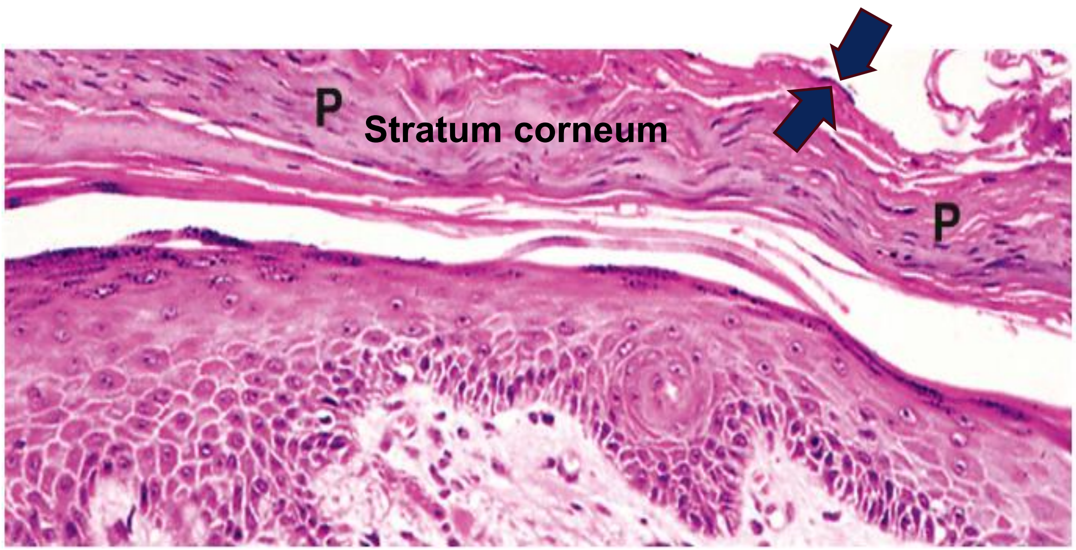
Stratum Basal



https://www.goodrx.com/conditions/chickenpox/chickenpox-in-black-skin

Definition of Terms

- **Parakeratosis** – retention of nuclei in the keratinocytes of the **stratum corneum**
  - Thining or loss of granular layer
  - Increased cell turnover



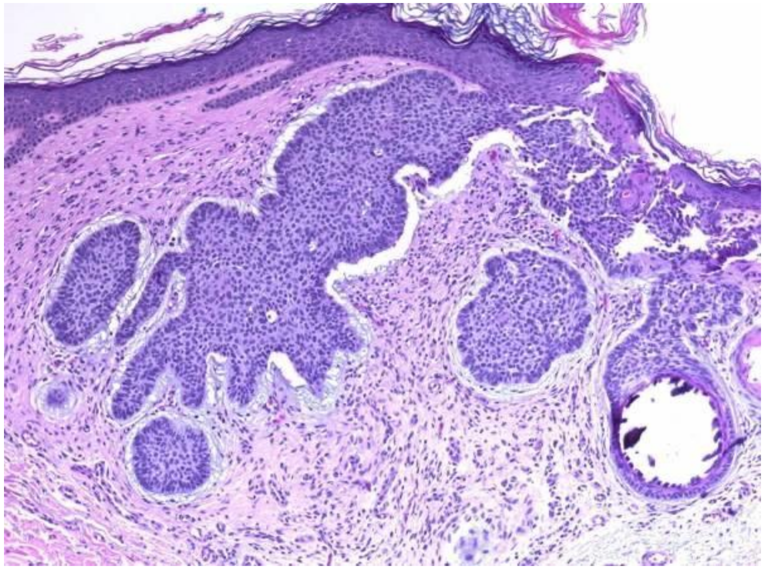
From O'Dowd G, Bell S, Wright S: Wheater's Pathology, 6e, 2020, Fig. 21.8, Elsevier,

- Associated conditions: Psoriasis, Actinic keratosis, Dandruff

Conditions Affecting Skin: Neoplasms

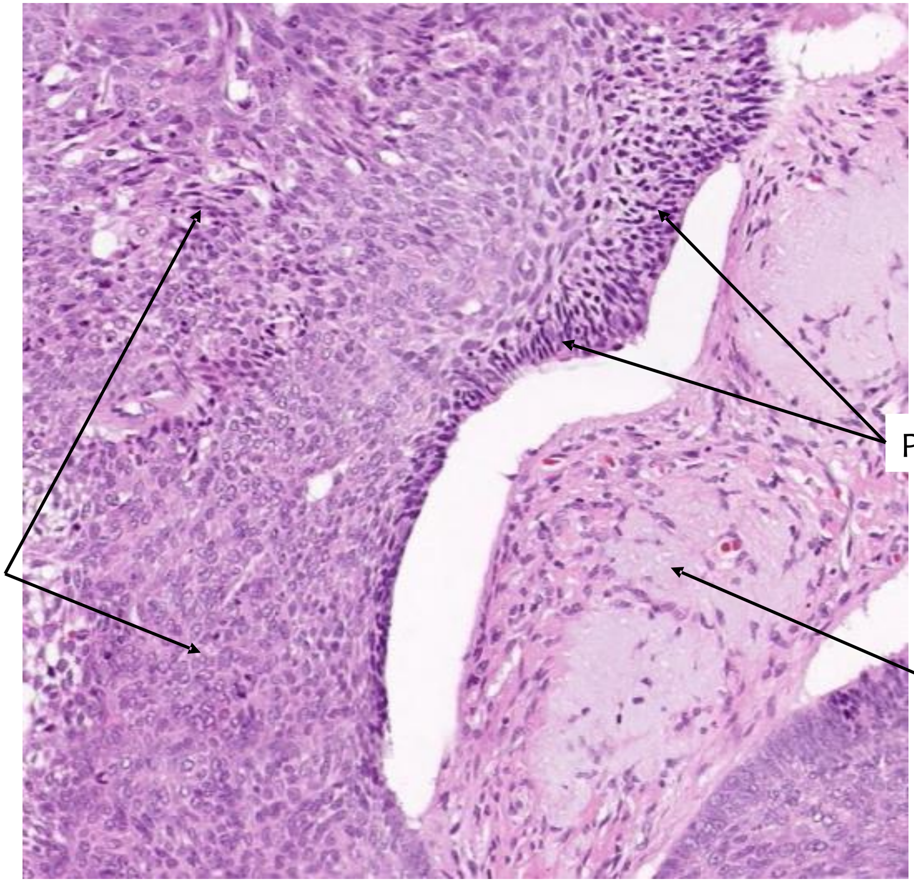
Basal cell carcinoma

- Predisposing Factor: **UV light**
- Pathophysiology: **Proliferation of basal stem cells**
- Dark nuclei with sparse, poorly defined cytoplasm
- Cells at the periphery have a characteristic palisaded appearance
- Central cells are more randomly arranged
- Invades the dermis and deeper-lying structures
- Does not typically metastasize



Basal cell carcinoma

May undergo ulceration



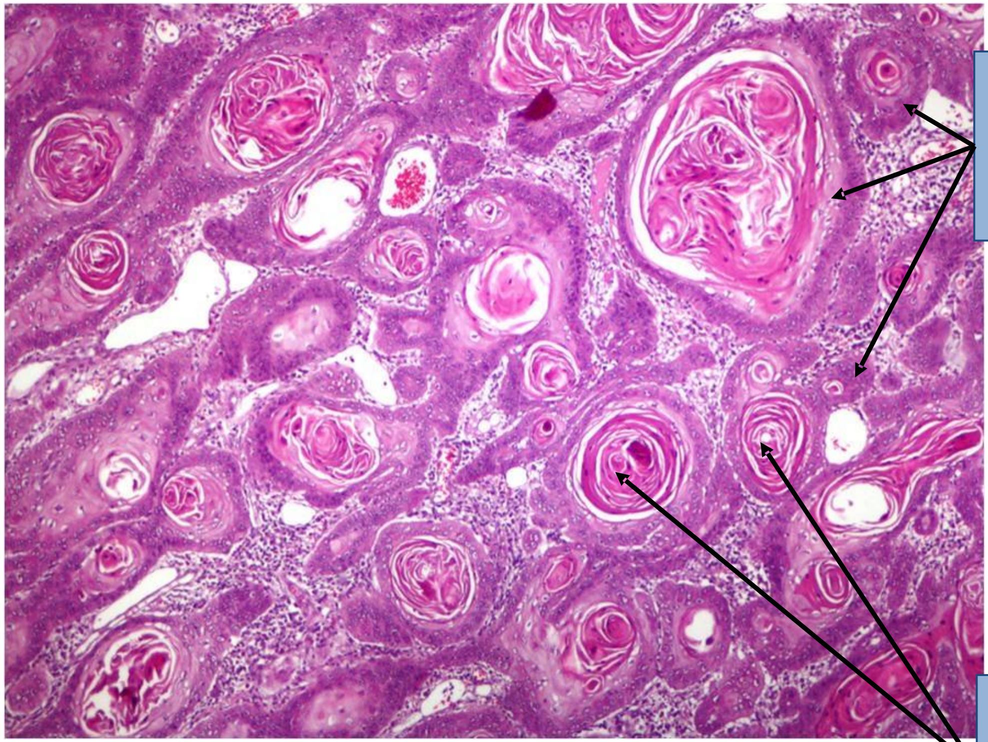
Squamous cell carcinoma

- **Malignant tumor of keratinocytes**
- Exposure to UV with DNA damage (inactivation of P53 gene)
- Loss of orderly maturation with variability in nuclear size and shape
- **Hyperkeratosis and parakeratosis**
- Common in elderly (> 70 years)
- Fair-skinned > Dark-skinned
- Predisposing factors :
  - Sunlight (UV)
  - Industrial carcinogens
  - Chronic ulcers
  - Tobacco chewing
  - Burns
  - Ionizing radiation
  - Betel nut chewing
  - Arsenic exposure

- **More common in the head and neck region (likely due to excessive sunlight exposure)**

Squamous cell carcinoma

Dermis deeply infiltrated by islands and sheets of malignant squamous cells.



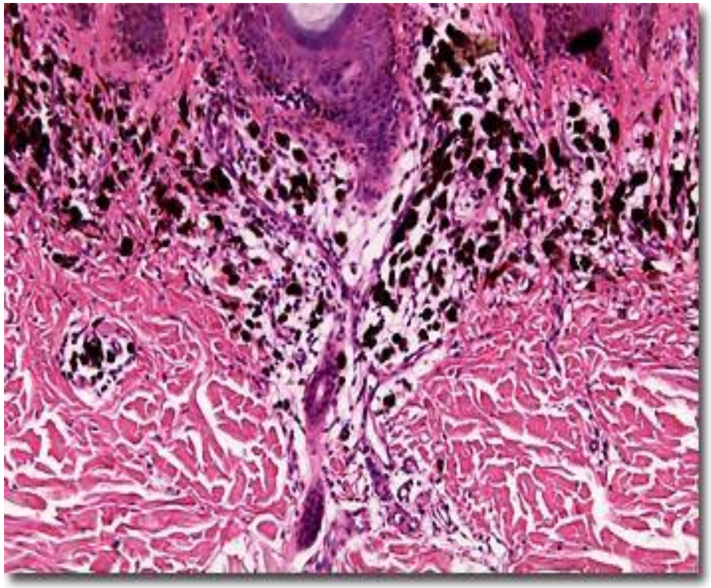
Koyuncuer, Ali. (2014). Histopathological evaluation of non-melanoma skin cancer. World journal of surgical oncology. 12. 159. 10.1186/1477-7819-12-159.

The islands have undifferentiated cells, resembling basal cells, around the perimeter.

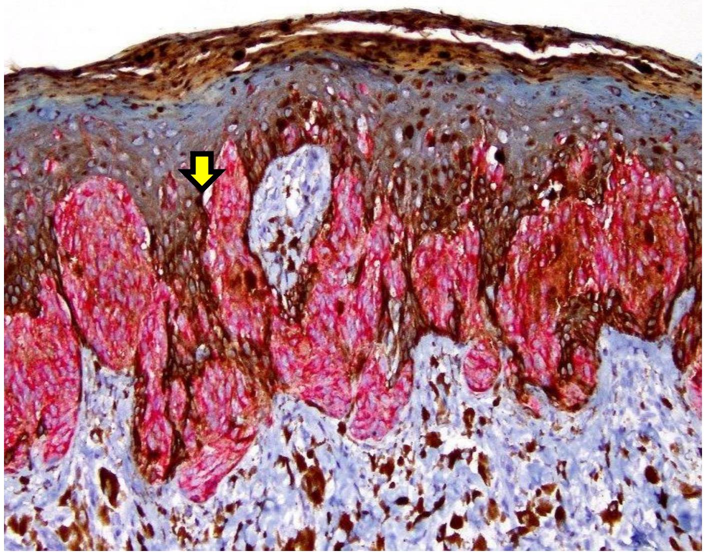
Islands show squamous differentiation with formation of squamous **pearls or swirls.**

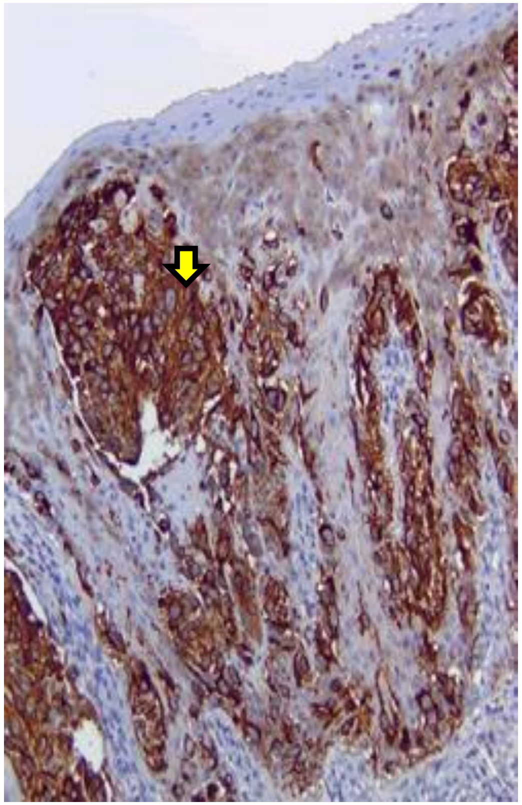
Malignant Melanoma

- Malignant transformation of melanocytes.
- Causes the highest number of skin cancer-related deaths in the US
- Predisposing Factor: **UV light**
- Increased number of melanocytes with large atypical morphology -- arranged at the dermo-epidermal junction
- Asymmetry - Border - Color - Diameter – Evolving (ABCDE)
- May invade the dermis
- Metastases (May be fatal)



Malignant Melanoma





Melan A -immunohistochemistry marker, a specific antibody that stains malignant melanoma cells

Conditions Affecting Skin: Pigmentation

Albinism

- Genetic (autosomal recessive) mutations affecting melanocytes
- Loss of pigmentation of the skin, hair, and eyes
- Lack of tyrosinase
- Types: Ocular, Oculocutaneous
- Long-term implications
  - Skin cancers
  - Reduced visual acuity/photophobia (macular hypoplasia)
  - Social stigma

Vitiligo

- Auto-immune depigmentation disorder
- Destruction of melanocytes
- Types: **Focal, Segmental, Generalized**
- Treatment:
    - Medical
        - Topical steroid therapy
        - Photochemotherapy
        - Depigmentation
    - Surgical
        - Autologous skin graft
        - Micropigmentation
        - Melanocyte transplants

Psoriasis

- A chronic inflammatory and hyperproliferative disorder of the skin
- Clinically manifested as well-circumscribed, erythematous, and itchy plaques covered with silvery scales (knees, elbows, lower back, scalp)
- Epidermal turnover averages 8-10 days
- Characterized by **hyperkeratosis** and **parakeratosis**
- **Predisposing factors:**
  - Genetic background
  - Inciting factors (stress)
- **Pathogenesis:**
  - Exogenous/Endogenous antigens.
  - Antigen presentation by APCs (Langerhans cells)
  - T-lymphocyte-mediated immune response
  - Cytokine secretion
  - Inflammation & cellular hyperproliferation
  - Clinical lesions of psoriasis

Psoriasis

- Predisposing factors
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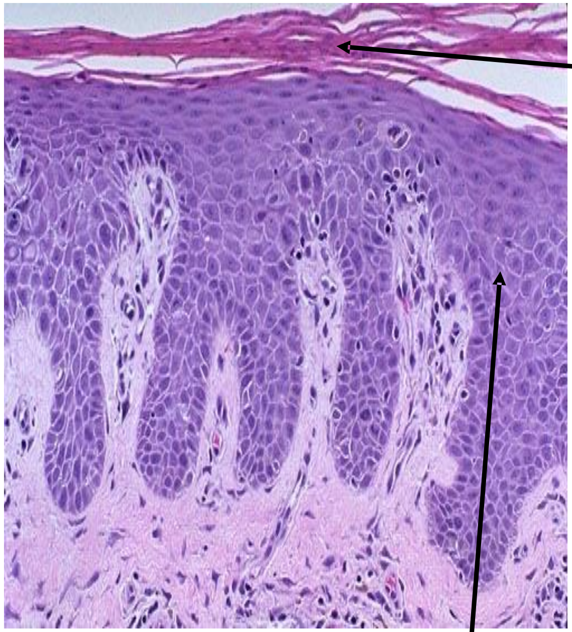
Psoriasis

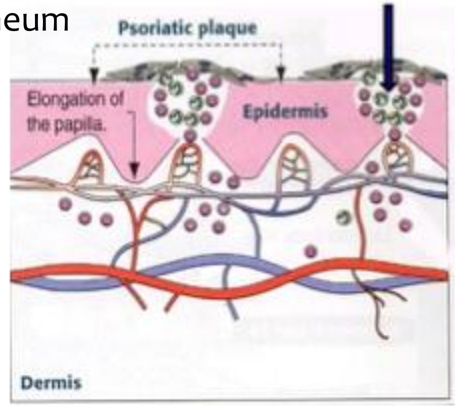


Parakeratosis:
Retention of nuclei
in the stratum corneum

Epidermal hyperplasia
Increased rate of proliferation

Prominent itchy areas with increased skin scaling and peeling

Munro's micro abscess

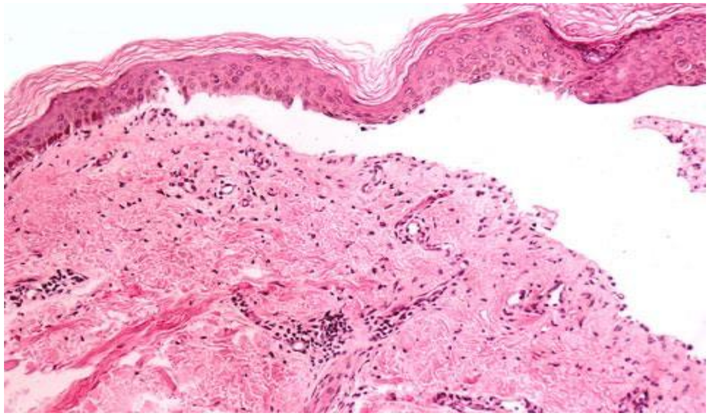


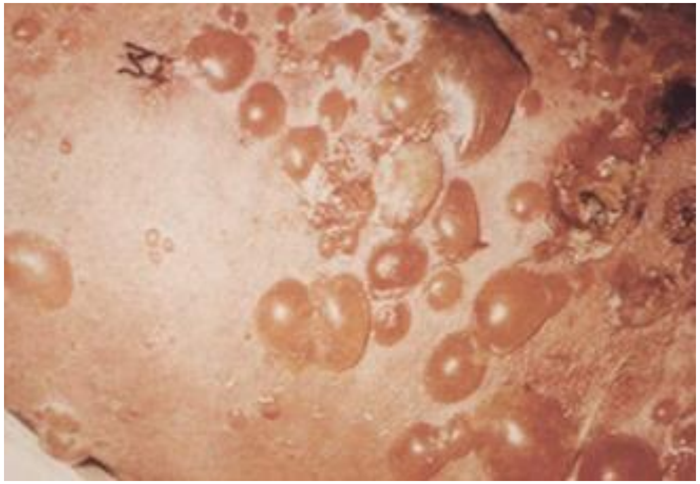
This leads to the shedding of the epidermis constantly, resulting in scales seen as whitish patches

Conditions Affecting Skin: Epidermis and Appendages

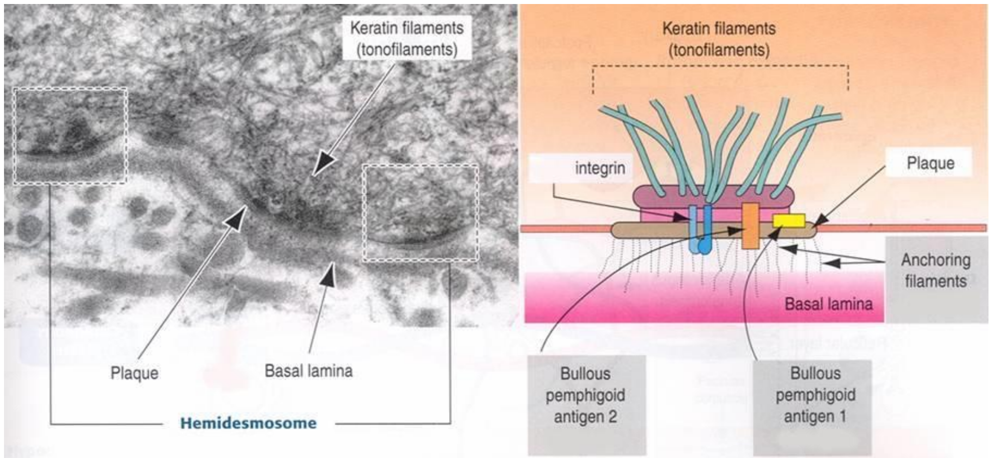
Bullous Pemphigoid

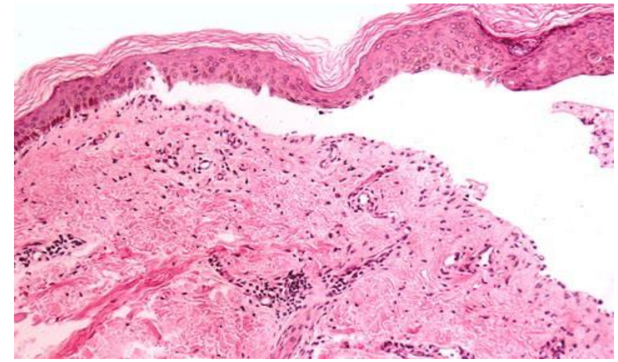
- Damage to hemidesmosomes – separation of epidermis from dermis
- Chronic autoimmune blistering disease: skin and mucous membranes (bulla large fluid filled vesicles)
- Antibodies (IgG) specific to hemidesmosomes bind to basement membrane and stimulate leukocytic infiltration
- Eosinophils release proteases that degrade hemidesmosomes
- Fluid accumulation – blister formation
- Characteristic by large blisters that don't easily rupture



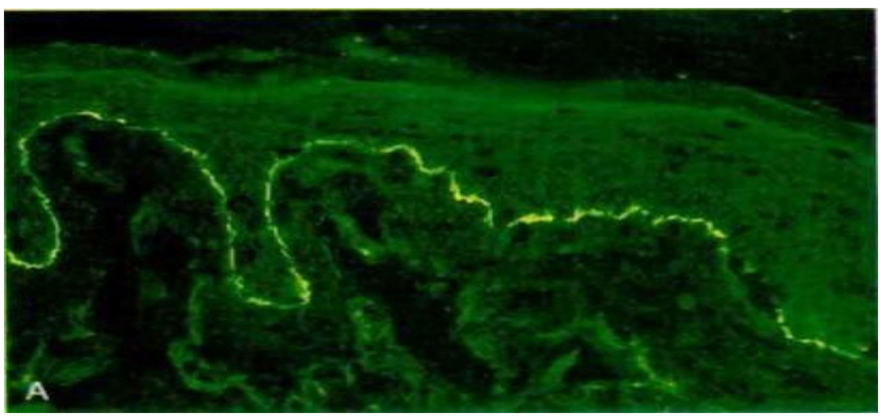


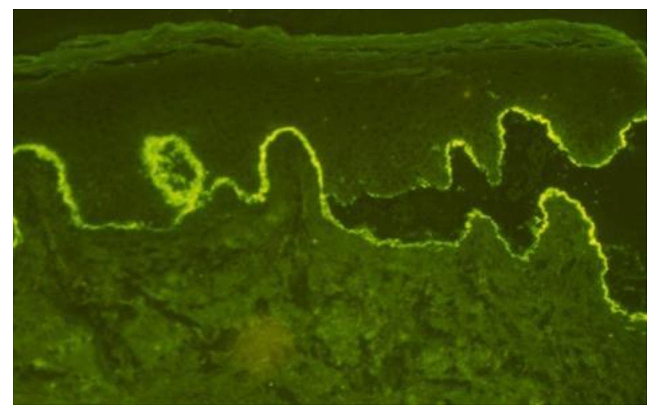
Bullous Pemphigoid





Bullous pemphigoid

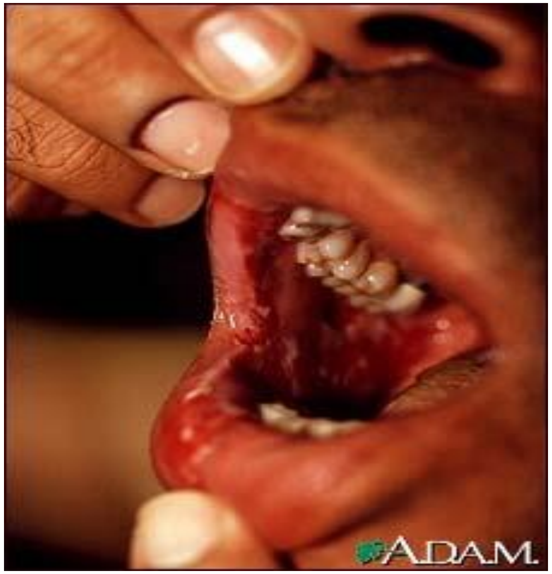


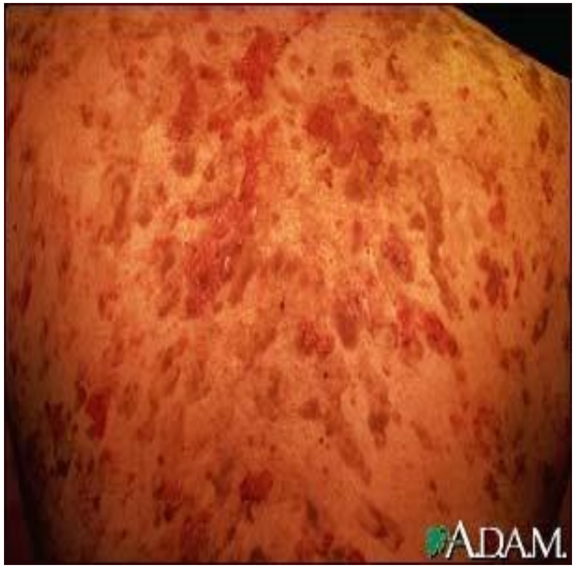


Linear deposition of Ig & complement in the Basement Membrane

Pemphigus Vulgaris

- Pemphix (blister/bubble)
- Rare autoimmune disorder affecting the epidermis and mucosal epithelium
- Desmosomes – separation of the stratum spinosum keratinocytes one from another
- Antibodies target cadherins and desmoplakins and desmogleins
- Atrophy of the prickle cell layer
- Blister formation: easy to rupture – Nikolsky's sign, skin shears off easily when wiped





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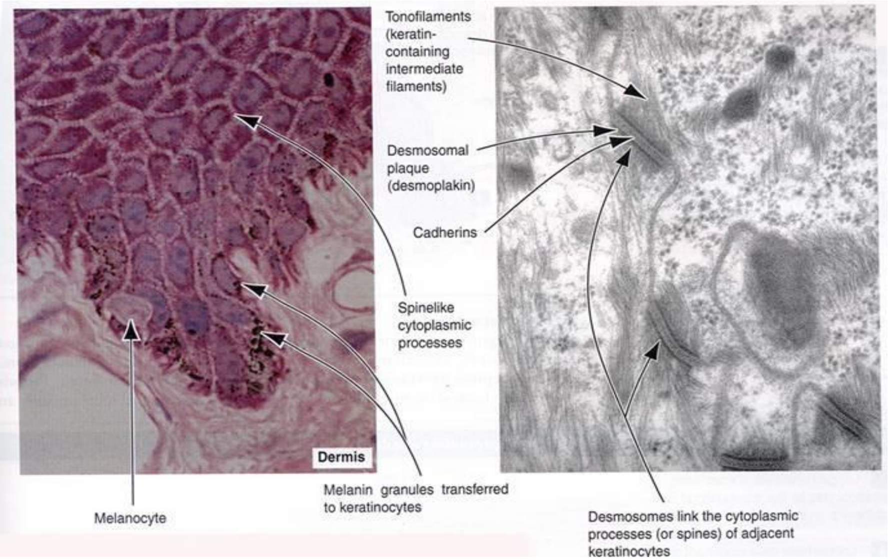
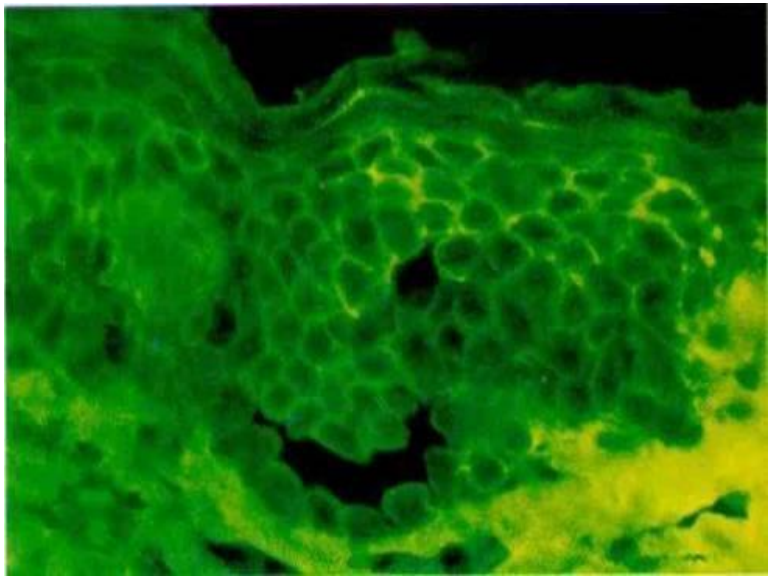
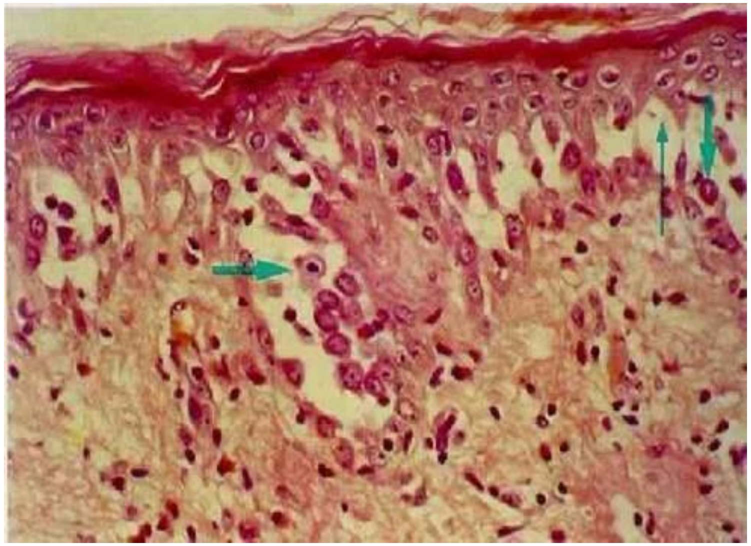
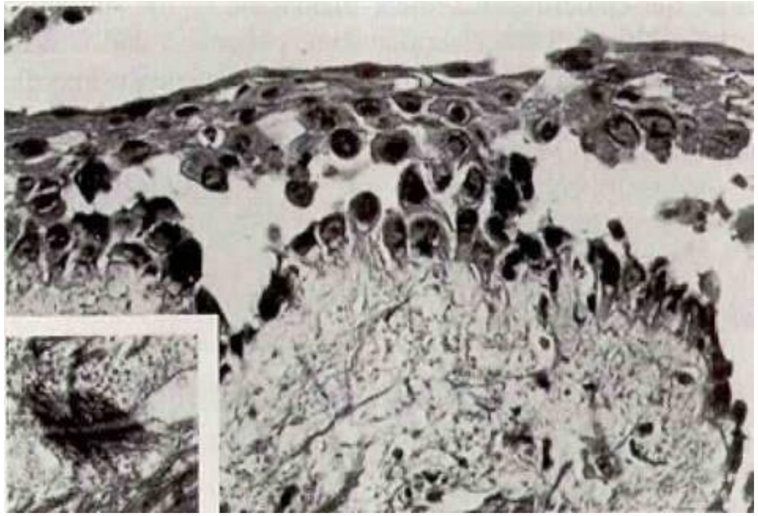
Grenada, West Indies

Pemphigus Vulgaris



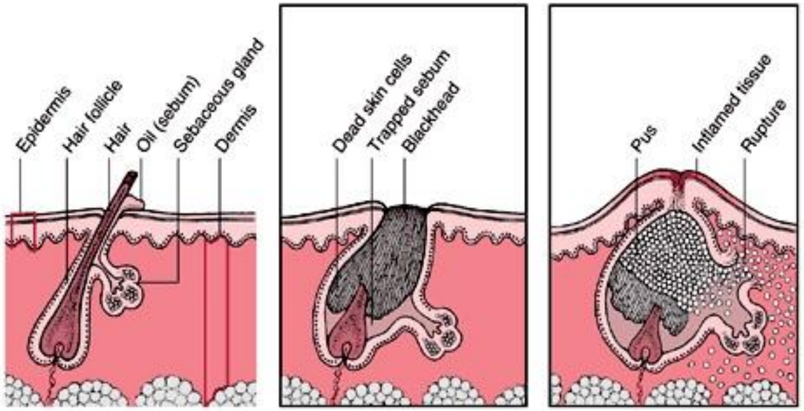






ACNE- inflammation of a sebaceous gland

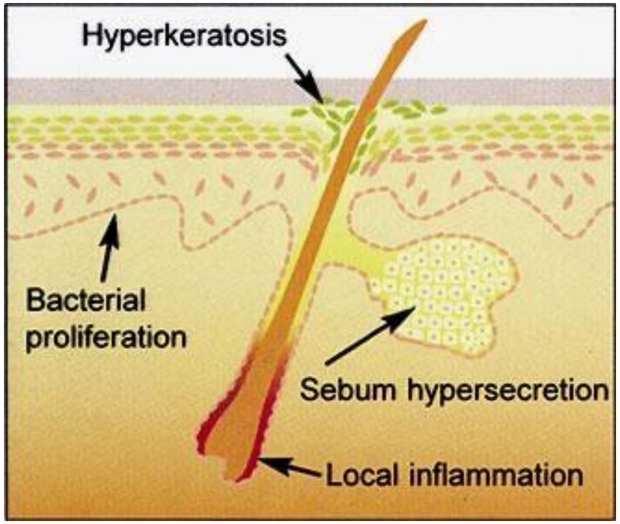
- Affects 85 – 100% of people at some point in life
- Genetic predisposition
- Common in adolescents (prior to the onset of puberty - **adrenal androgens**): in all age groups
- Affects face, back, chest upper arm – areas with increased sebaceous glands



Cross Section of Normal Skin

Mild Acne

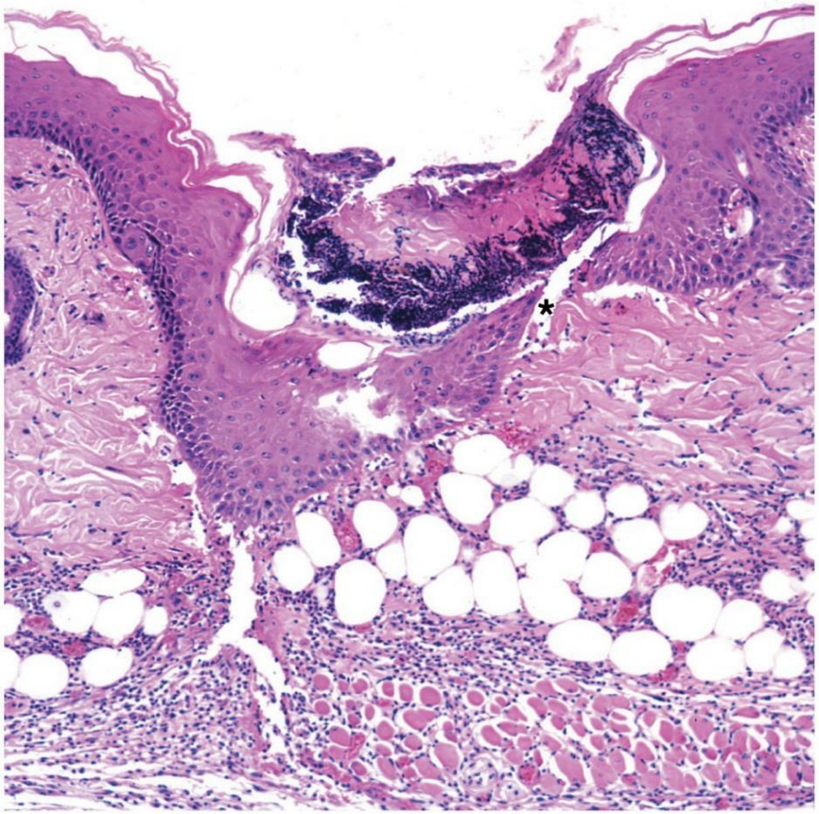
Severe Acne



Tissue Repair: Wound Healing

Types of Wound Healing

- Skin damage sets in motion a sequence of events that repairs the skin to its normal (or near normal) structure and function.
- 2 types:
  - Epidermal Wound healing
  - Deep wound healing
- Epidermal wound healing:
  - Occurs following wounds that affect only the epidermis
- Deep wound healing:
  - Occurs following wounds that penetrate the dermis.

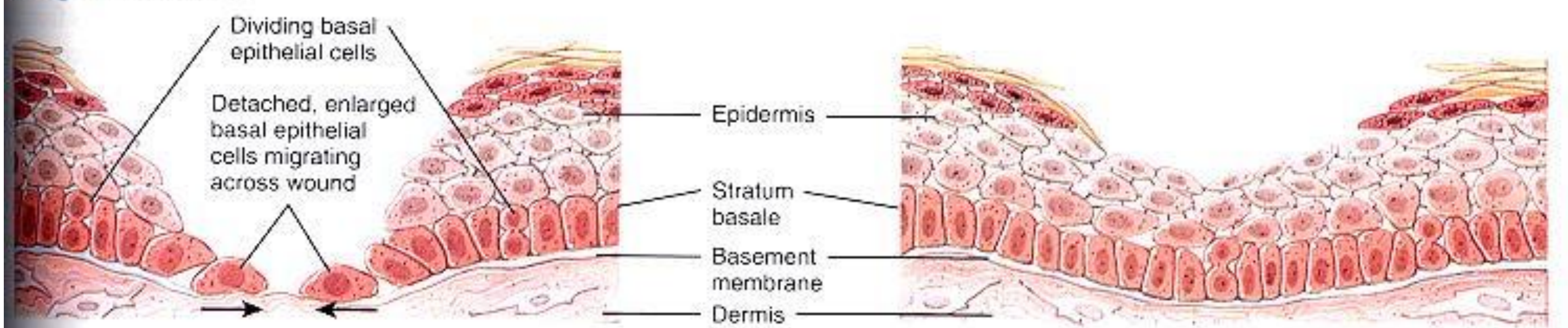


Epidermal Wound Healing

- In response to an injury, the basal cells of the epidermis surrounding the wound, break contact with the basement membrane, enlarge and migrate across the wound.
- The cells migrate from opposite sides until they meet (in a sheet-like formation).
- Epidermal growth factor stimulates basal stem cells to divide and replace the lost cells, that have moved into the wound.
- These new cells divide to thicken the new epithelium.

Figure 5.6 Skin wound healing.

In an epidermal wound, the injury is restricted to the epidermis; in a deep wound, the injury extends deep into the dermis.



(a) Division of basal epithelial cells and migration across wound

(b) Thickening of epidermis

Epidermal wound healing

Deep Wound healing

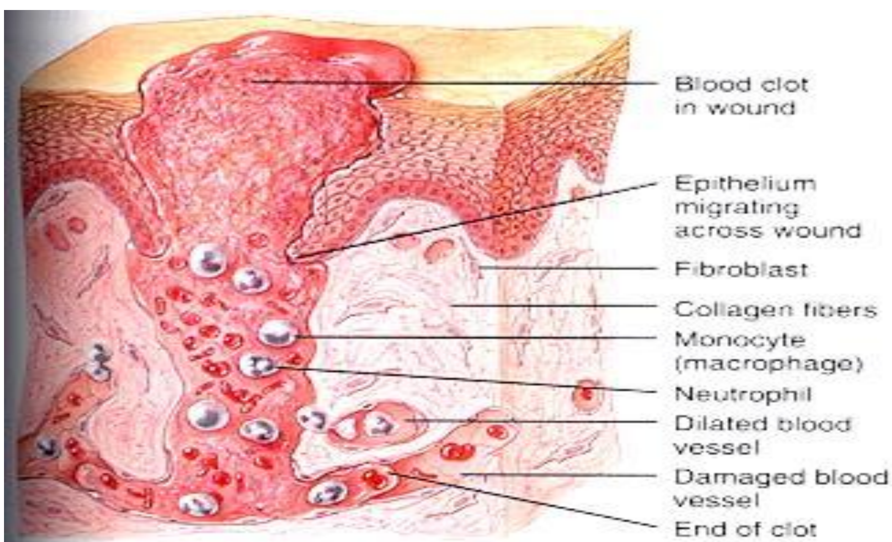
- As the injury involves multiple tissue layers, the healing process is more complex.
- Healing involves scar formation and can include loss of function.
- This type of healing occurs in 4 phases:
  1. Hemostasis Phase (initial injury to hours after wound)
  2. Inflammatory Phase (up to about 4 days after wound)
  3. Proliferative Phase (from the 3rd day to weeks after the wound) (type III collagen dominant)
  4. Maturation Phase (up to 6+ months after wound) (Type III collagen replaced with Type I)

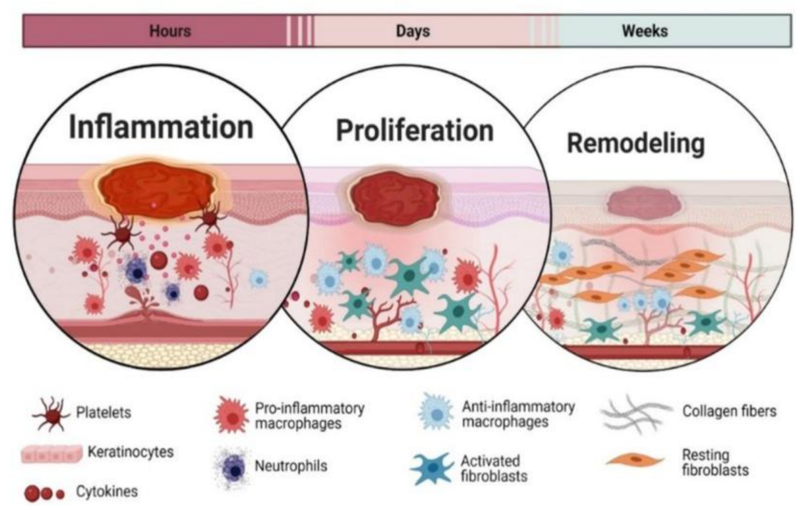
1. Hemostasis Phase

This phase helps to stop any bleeding, serve as an initial barrier against infection and lay the foundation for the inflammation phase.

Key components of this phase:

- **Vasoconstriction**: Blood vessels constrict to reduce blood flow and minimize blood loss.
- **Platelet Aggregation**: Platelets adhere to the site of injury and aggregate to form a clot.
- **Coagulation Cascade**: Various proteins in the blood work together to strengthen the platelets and form a stable clot.

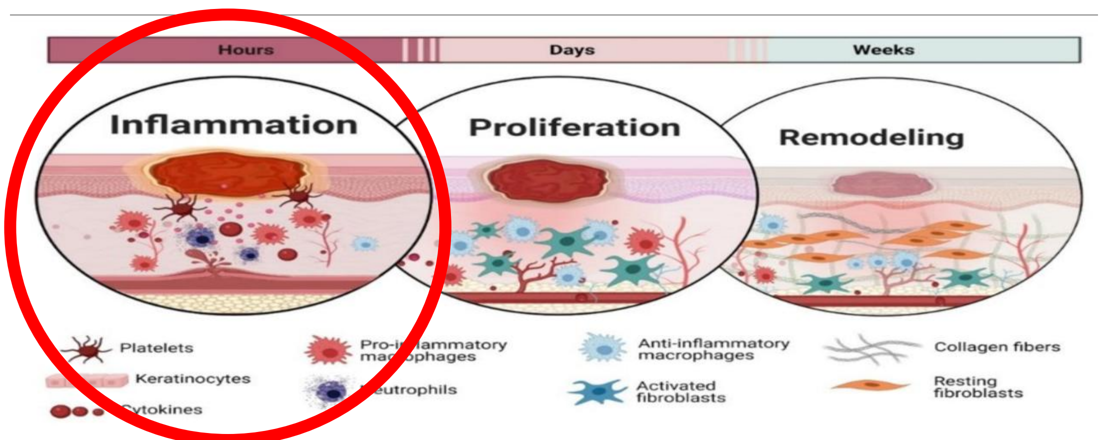




2. Inflammatory Phase

Inflammatory white blood cells move to the wound to help, remove damaged cells and bacteria, and create a scab to cover the injured area. These white blood cells, growth factors, nutrients and enzymes create the swelling, heat, pain and redness commonly seen during this stage of wound healing

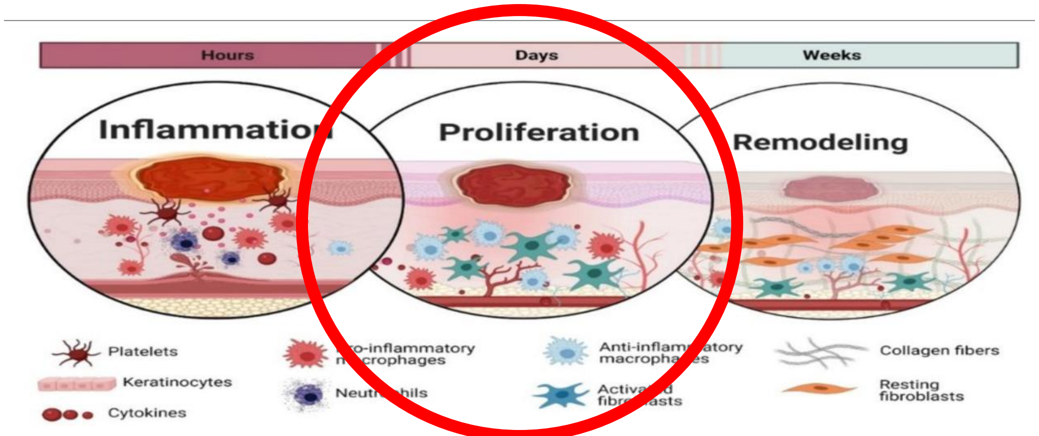
- Key components of this phase:
- **Vasodilation:** Blood vessels expand, increasing blood flow to the area to facilitate the delivery of immune cells.
- **Immune Cell Activation:** White blood cells such as neutrophils and macrophages arrive at the wound site to combat potential pathogens.
- **Response to Injury:** Cytokines and growth factors are released, signaling the next phase of healing.



3. Proliferative Phase

The proliferative phase of wound healing is when the wound is rebuilt with new tissue made up of collagen and extracellular matrix. Myofibroblasts cause the wound to contract by gripping the wound edges and pulling them together using a mechanism similar to that of smooth muscle cells.

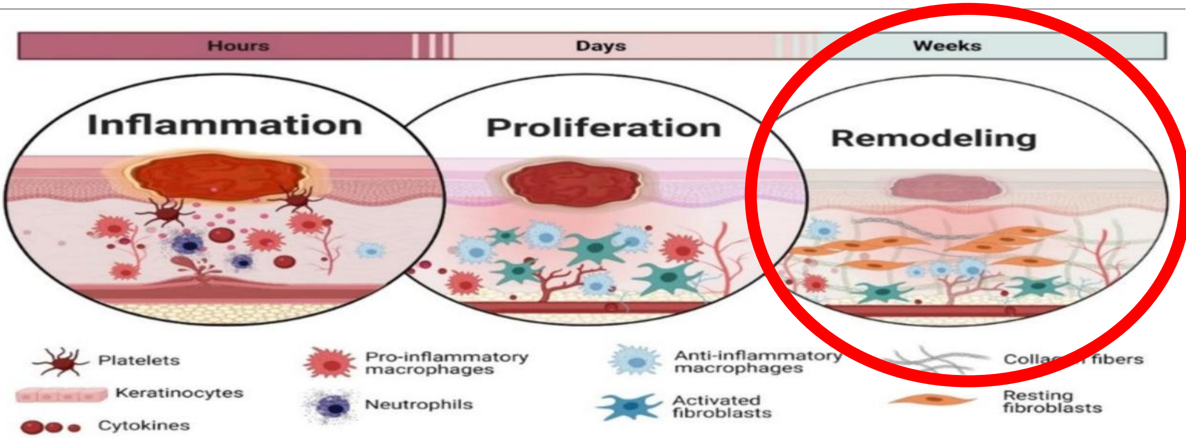
- Key components of this phase:
- **Angiogenesis:** Formation of new blood vessels to supply nutrients and oxygen to the healing tissue.
- **Fibroplasia:** Fibroblasts synthesize collagen and extracellular matrix to form new connective tissue.
- **Epithelialization:** Skin cells migrate to cover the wound area, leading to closure and barrier restoration



4. Maturation Phase

The maturation phase is when collagen is remodeled from type III to type I and the wound fully closes. The cells that that are no longer needed are removed by apoptosis, or programmed cell death.

- Key components of this phase:
- **Collagen Remodeling:** The collagen matrix is reorganized, cross-linked, and refined to improve tissue tensile strength.
- **Scar Formation:** Over time, scars may become less visible as tissue matures.
- **Restoration of Function:** The healed area gradually returns to normal function, although it may never completely regain its original state.



Abnormal Wound Healing

- **Hypertrophic scar** - raised more than normal, but within original wound boundary Mainly type III collagen in well-organized rows.
- **Keloid scar** - in excess of the boundary, and extending into surrounding tissue
  - Mainly highly disorganized types I and III collagen
  - More common in persons who are darker skinned
  - Common locations – ear, neck, sternum, upper extremities

Hypertrophic scar

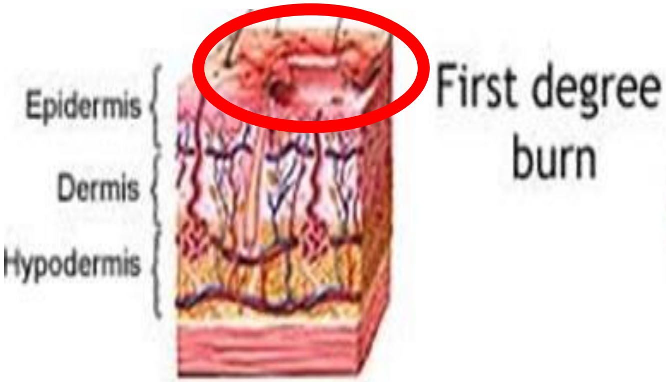
Keloids (Locally known as plumbs)
Cause unknown

Tissue Repair: Burns

Classification of Burns

First-Degree Burns (Superficial Burn)

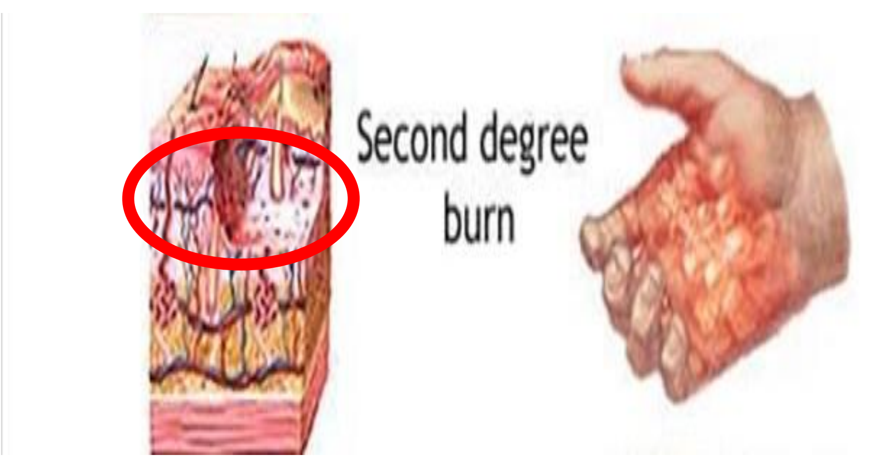
- **Affects**: Only the outer layer of the skin- partial thickness epidermis.
- **Symptoms**: Redness, mild swelling, pain, and dryness without blisters. Heals spontaneously
- **Example**: Mild sunburn



Classification of Burns

Second-Degree Burns

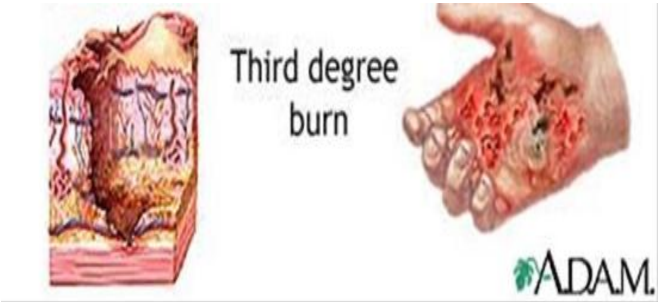
- **Affects**: Both the epidermis and the underlying dermis. Epithelial water barrier disrupted.
- **Symptoms**: Redness, swelling, pain, and blistering. Heals spontaneously.
- **Example**: Scalds from hot liquids

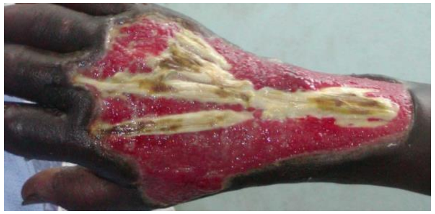


Classification of Burns

Third-Degree Burns

- **Affects**: These burns destroy both the epidermis and dermis, and may extend into the subcutaneous tissue.Water barrier disrupted, nerves and blood vessels destroyed, does not heal spontaneously, fluid loss is extensive
- **Symptoms**: Redness, swelling, pain, and blistering. Heals spontaneously.
- **Example**: fire/flame, electrical, chemical, prolonged contact with hot objects



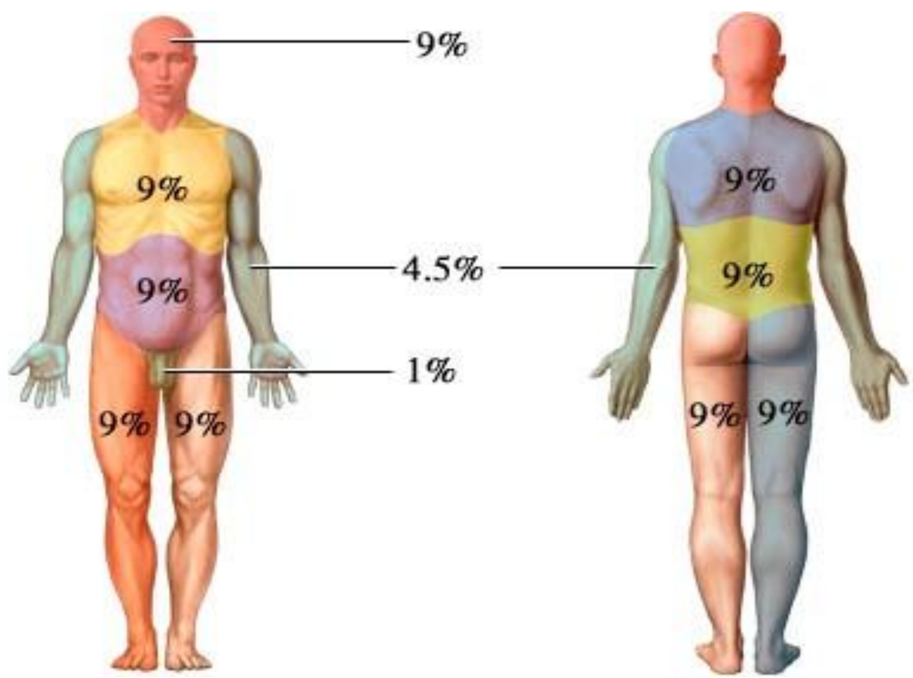


Burns

Wallace Rule of Nines

Technique: estimate the total body surface area (TBSA) of a burn patient.

- Helps assess severity of burns
- Estimates fluid resuscitation requirements
- Provides data to estimate patient outcome



https://www.health.state.mn.us/communities/ep/surge/burn/tbsa.html

Burns

Classification:

1. **The entire head:** 9% (4.5% for anterior and posterior)
2. **The entire trunk:** 36% (can be further broken down into 18% for the anterior torso and 18% for the posterior torso)
3. **The anterior aspect of the trunk:** Can further be divided into the chest and abdomen, each representing 9%
4. **The upper extremities:** 18% (includes 9% for each upper extremity)
5. **Each upper extremity:** Can be further divided into its respective anterior and posterior portions (4.5% each).
6. **The lower extremities:** 36% (18% for each); can be further divided into 9% for the anterior and 9% for the posterior aspect
7. **The groin:** 1%